

Milan J. Hancock

Indiana, USA

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GitHub: github.com/milanjh56/projects | Portfolio: milanjh56.github.io

Education

Indiana University Bloomington - B.S. Intelligent Systems Engineering (*Expected May 2028*)

Dean's List: Fall 2024 - Spring 2026 | 4.00/4.00 GPA

Skills

Languages: C/C++, Python, SystemVerilog

Machine Learning: Google Colab, Scikit-learn, Tensorflow

CAD: Fusion360, OnShape, LTSpice

Hardware/Embedded Systems: Arduino, Raspberry Pi, soldering

Tools: Linux, Git, CLI, JetStream2 virtual machine

Projects

(In progress) Real-Time Driving Event Classification using BeamNG.tech

- Developing real-time machine learning pipeline to classify driving events from simulated IMU data.
- Writing Python script that interfaces with BeamNG.tech to automate data collection and stream accelerometer and gyroscope measurements at specified sampling rates.
- Training and evaluating classification models for live inference on sliding-window sensor data.

DIY Sim Racing Shifter and Handbrake

- Built custom USB sequential shifter and analog handbrake that share a single Arduino Pro Micro (ATmega32U4).
 - Designed and 3D-printed a desk-mounted shifter that integrates microswitches to achieve an immersive shifter feel.
 - Built a handbrake assembly using scrap wood, creating custom lever mechanisms, pivots, and adjustable linkages.
 - Repurposed a thumb joystick module as an analog position sensor for the handbrake, utilizing a mechanical linkage and software calibration to convert partial joystick travel into a full-range analog brake axis.
 - Developed firmware in C++ (Arduino IDE) to process microswitch inputs and analog joystick readings.
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Activities

Aerial Robotics Club (Fall 2025)

- Participated in team drone design, Arduino integration, and iterative prototype development.
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Work Experience

Cascades Golf Course: Laborer II (May 2025 - Present)

- Ensure course playability by mowing, changing pins, maintaining irrigation systems, raking bunkers, trimming trees.

TurfPro Plus: Groundskeeper (May 2024 - August 2024)

- Maintained 24-acre property by mowing, trimming trees, fertilization, irrigation, and sod installation.

Sam's Club: Member Assist (May 2023 - August 2023)

- Assisted shoppers through cart retrieval and loading groceries while maintaining an organized parking lot.